



COMSATS University Islamabad, Vehari Campus

Project Proposal
(SCOPE DOCUMENT)

For
Alzheimer's Disease Prediction Using Deep
Learning Techniques

By

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SCOPE DOCUMENT REVISION HISTORY

No.	Comment	Action

Supervisor Signature:_____

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Abstract

Alzheimer's Disease is a brain disease that slowly destroys memory and thinking. Early diagnosis is very important for slowing the disease's growth and giving patients better care. But the old ways of figuring out what illness someone has take a long time, cost a lot of money, and often find it later. Our software will try to fix this by using deep learning to find out who has Alzheimer's Disease. The algorithm will look for patterns in medical data, like brain scans, that are linked to Alzheimer's disease in its early stages.

1. Introduction

Alzheimer's Disease is a progressive neurodegenerative disorder that significantly impacts memory, cognition, and daily functioning. Early and accurate diagnosis is critical for timely intervention and care planning. Magnetic Resonance Imaging plays an essential role in the diagnostic process by revealing structural brain changes associated with AD, such as hippocampal atrophy and cortical thinning. However, manual interpretation of MRI scans is often time-consuming, subjective, and prone to variability among experts.

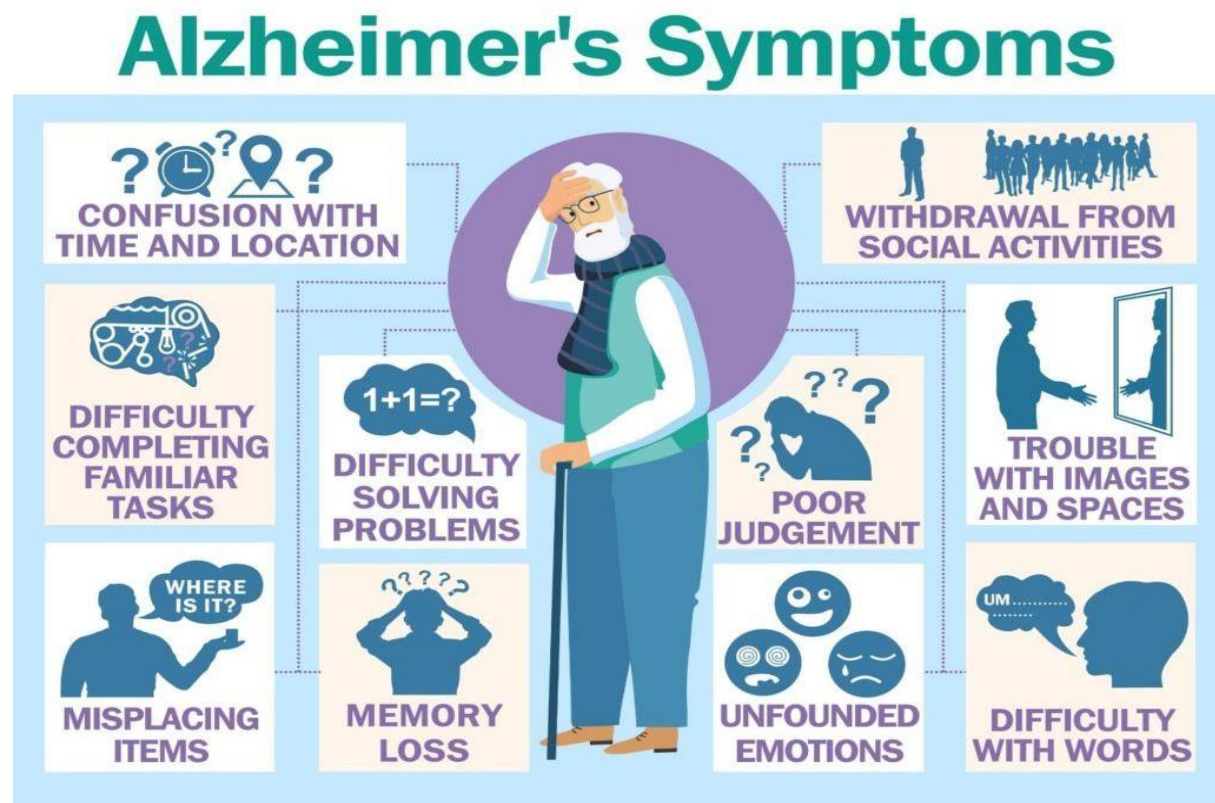
Advancements in deep learning, particularly Convolutional Neural Networks, offer promising solutions for improving the accuracy and efficiency of medical image analysis. These models can learn complex patterns from imaging data, enabling automated and consistent diagnosis support.

This project will aim to develop a CNN-based system for predicting Alzheimer's Disease from MRI scans. By training the model on a large, labelled dataset, we will seek to detect subtle biomarkers associated with AD. To enhance the model's generalization, we will employ robust data augmentation techniques. To ensure practical utility, we will also design an accessible interface that will allow healthcare professionals to upload MRI scans and receive predictive insights with visual explanations. This tool will be intended to assist, not replace, clinical decision-making, supporting ethical and informed use in medical settings. Our goal will be to contribute to more timely, objective, and equitable diagnostic support for Alzheimer's Disease through the responsible integration of deep learning technologies in healthcare.

2. Problem Statement

Alzheimer's Disease is a neurological illness that makes memory and thinking deteriorate over time. Early diagnosis is particularly crucial for slowing its growth and offering patients better care. But the old means of finding out what sickness someone has taken a long time, cost a lot of money, and often detect it later. Our software will try to fix this by applying deep learning to figure out who has Alzheimer's Disease. The algorithm will seek for trends in medical data, including brain scans, that are connected to early-stage Alzheimer's.

We will be developing this system to support medical professionals in making faster and more accurate diagnoses. Although similar systems will exist, re-implementing one will help us better understand how artificial intelligence is applied in healthcare. This approach will allow us to experiment with different model architectures and improve upon existing solutions. Through this project, we will expect to learn Python programming, machine learning principles, and deep learning frameworks like TensorFlow or PyTorch. We will also gain experience in data preprocessing, model training, and evaluating prediction performance. Handling medical datasets will teach us data management and ethical considerations. By applying these techniques, we will understand the real-world impact of AI in medicine. Overall, this project will enhance both our technical and analytical skills. It will also prepare us for future research or careers in AI-based healthcare systems.



Alzheimer's Symptoms Description:

This image highlights the common symptoms of Alzheimer's disease in a visually engaging way. It shows an elderly person surrounded by signs such as memory loss, confusion with time and location, difficulty completing familiar tasks, and poor judgment. Other symptoms include trouble with words, misplacing items, withdrawal from social activities, and difficulty solving problems. Emotional and perceptual changes like unfounded emotions and trouble with images and spaces are also depicted. This infographic serves as a quick reference to recognize early warning signs of Alzheimer's.

3. Problem Solution for Proposed System

We want to use deep learning to look at medical data, like MRI brain scans, to see if someone has Alzheimer's Disease early on. Convolutional Neural Networks and other deep learning models are good at finding patterns in image data that the human eye can't see. By training the model on labeled datasets, the system will learn to sort brain images into groups like "healthy," "mild cognitive impairment," or "Alzheimer's." This helps doctors diagnose patients sooner and more accurately than other methods.

The system will be created on open-source Alzheimer's datasets, which will make it easy to use and scale. After it has been trained, the model will take new patient data as input and give a prediction about how likely it is that the patient has Alzheimer's. The prediction can help doctors make decisions more quickly and with more confidence.

To make the system user-friendly, we will plan to integrate a simple graphical interface where medical staff can upload patient scans and receive real-time predictions. Additionally, our system will include model evaluation metrics like accuracy, precision, recall, and a confusion matrix to ensure reliability. We will also explore data preprocessing techniques like normalization and augmentation to improve model performance. The system will be tested and validated to ensure it performs well on unseen data.

This project will provide a low-cost, non-invasive tool that can be used in hospitals and clinics to support early diagnosis. It addresses the problem of delayed detection by automating the analysis process using AI. Our solution contributes not only to patient care but also to our learning in AI, deep learning, and healthcare applications.

4. Related System Analysis

Deep AD:

Deep AD is a CNN-based model that classifies MRI images into Alzheimer's and non-Alzheimer's categories. It achieves high accuracy by using preprocessed brain scans and multiple convolutional layers. The system is fully automated and demonstrates the potential of deep learning in early diagnosis. However, it lacks a user-friendly interface for clinical use.

VGG-16 Based Alzheimer Detection System: This system uses a pre-trained VGG-16 model to extract features from MRI images for Alzheimer’s prediction. It benefits from transfer learning, which reduces training time and increases accuracy. The model is effective for image classification but requires large computational resources. It does not offer real-time prediction capabilities.

3D CNN for Alzheimer's Classification Using MRI Data: This model uses a 3D Convolutional Neural Network to analyze volumetric MRI data, capturing spatial features more effectively. It provides better accuracy than 2D CNNs for medical imaging tasks. However, it demands high memory and processing power, making it less practical for real-time clinical deployment. The system is mainly used for academic research.

Related System Analysis: These current systems show that deep learning can make Alzheimer's diagnoses much more reliable. But healthcare providers can't easily access, use, or apply most of them in real time. Our proposed project aims to fill in these gaps by creating a lightweight, easy-to-use deep learning model that can be used in real-time clinical settings. This contribution will make things easier to use, lower costs, and help find problems early in places where resources are limited.

Table 1: Related System Analysis for the proposed project system.

Application Name	Limitations	Proposed Solution
VGG-16 Based Alzheimer Detection System	Requires high computational power for training and inference, and lacks scalability for low-resource environments.	We will design a lightweight deep learning model that balances accuracy and efficiency, making it suitable for deployment in resource-constrained settings such as rural hospitals or clinics.
3D CNN for Alzheimer’s Classification Using MRI Data	High memory usage and computational requirements limit its use to academic or research labs; it is not suitable for real-time or mobile use.	Our system will use 2D or optimised hybrid models to reduce computational load while maintaining reliable prediction accuracy, making it feasible for broader deployment.

5. Advantages of the Proposed System

Early Detection of Alzheimer's Disease

- Automation reduces human error
- Integrated patient and doctor management
- Secure medical data handling
- Scalable and efficient platform for large datasets
- Personalized recommendations improve healthcare access
- User-friendly interface for patients and professionals

6. Scope

This system facilitates early prediction of Alzheimer's using deep learning (CNNs). It allows MRI upload, returns disease stage predictions, and recommends doctors based on severity and location. It includes a chatbot for disease-related queries and secure user authentication. Modules will include image processing, prediction, user management, and reporting. The scope covers the complete flow from scan upload to result output, excluding manual diagnosis or offline support.

7. Modules

1. User Registration/Login
2. MRI Upload and Preprocessing
3. CNN Model for Prediction
4. Chatbot for Alzheimer's Info
5. Doctor Recommendation System
6. Admin Panel for User/Data Management

Module 1: User Registration/Login

This module allows users (patients, doctors, or researchers) to create secure accounts and access the system. New users will be able to register by providing basic information, while returning users can log in using their credentials. This ensures personalized access and protects sensitive medical data.

Module 2: MRI Upload and Preprocessing

Once logged in, users can upload their brain MRI scans in supported formats. The system will automatically perform preprocessing steps like resizing, normalization, and slice extraction to prepare the images for accurate analysis by the CNN model.

Module 3: CNN Model for Prediction

This is the core functionality of the system. After preprocessing, the uploaded MRI scan is passed through a trained 2D Convolutional Neural Network. The model will analyze the scan and predict whether signs of Alzheimer's Disease are present, showing results such as probability scores and highlighted brain regions using visual explanation tools.

Module 4: Chatbot for Alzheimer's Info

To assist users in understanding the disease, a built-in AI-powered chatbot will provide educational content about Alzheimer's. Users can ask questions related to symptoms, stages, prevention, treatment options, and lifestyle tips. It serves as a 24/7 support tool for general awareness and guidance.

Module 5: Doctor Recommendation System (Special Feature):

Based on the prediction result and the user's location or preferences, this module will recommend nearby certified neurologists or Alzheimer's specialists. It will provide contact details, available slots, and clinic addresses to help users schedule consultations for further evaluation or treatment.

Module 6: Admin Panel for User/Data Management

An admin-only module that allows system administrators to manage users, monitor uploaded data, view system logs, and maintain the overall health of the platform. It ensures proper functioning of the system, handles misuse prevention, and supports data privacy and security measures.

8. System Limitations

- Requires large labelled datasets
- Internet-dependent interface
- Medical data privacy and ethical handling

9. Software Process Methodology

Agile methodology will be used for iterative development, allowing frequent updates and user feedback. This suits our modular design and enhances flexibility in development.

10. Tools and Technologies:

Category	Name	Version	Purpose
Tool	Google Colab	3.10	Model Training
Technology	React.js	19.1.0	Frontend Framework
Technology	Keras	3.10.0	Neural Network Library
Technology	TensorFlow	2.19.0	Model Development
Technology	Django/Flask	5.2.4	Backend Framework
Tool	SQLite	3.50.2	Database

11. Project Stakeholders and Roles:

Sponsor	COMSATS University Islamabad, Vehari Campus
Stakeholder	Students: Designers, developers, and testers Project Supervisor: Academic mentor and technical guide FYP Committee: Evaluation and final approval authority

12. Team Members Individual Tasks/Work Division

Student Name	Role	Responsibility
Abu ul Hasnat	Backend	➤ MRI Upload (Module 2) ➤ CNN Model (Module 3) ➤ Doctor System (Module 5) ➤ Admin Pane (Module 6)

Fatima	Frontend	➤ User login (Module 1) ➤ Chatbot info (Module 4) ➤ Documentation
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13. Data Gathering Approach

We used research papers, publicly available datasets, and conducted informal discussions with doctors to gather functional requirements and understand the prediction process.

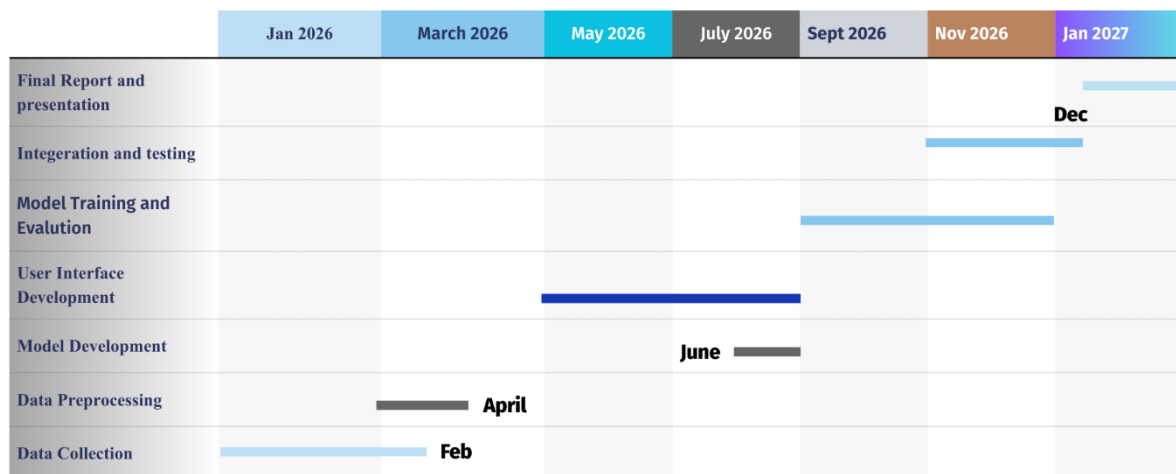
Website: <https://adni.loni.usc.edu/>

Website: <https://www.oasis-brains.org/>

14. Concepts

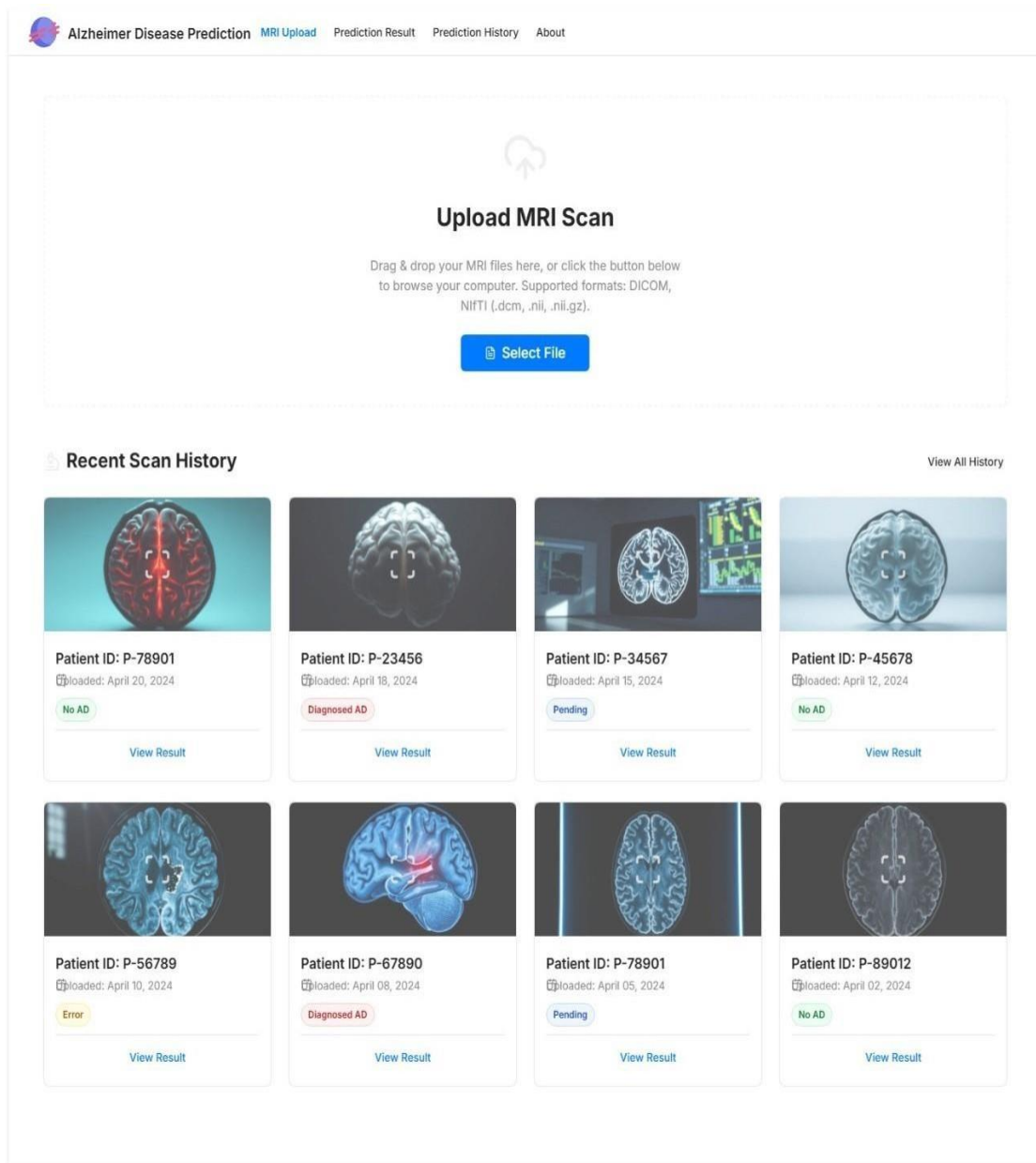
- **Convolutional Neural Networks:** Used for medical image classification.
- **Data Augmentation:** Increase dataset diversity for better training.
- **Collaborative Filtering:** Suggest doctors based on user behaviour.
- **Chatbot Systems:** NLP-based query resolution about Alzheimer's Disease.

15. Gantt chart:



16. Mockups:

MRI Scans:



About Project:

[Alzheimer Disease Prediction](#) [MRI Upload](#) [Prediction Result](#) [Prediction History](#) [About](#)

Our Mission & Vision

At Alzheimer Disease Prediction, our mission is to revolutionize early Alzheimer's diagnosis by providing medical professionals with an accurate, accessible, and user-friendly AI-powered tool. We believe that early detection is key to effective intervention and improving patient outcomes.

Our vision is a future where Alzheimer's disease can be identified with unprecedented precision, enabling timely medical support and research breakthroughs. We strive to be a trusted partner in the global fight against neurodegenerative diseases, fostering innovation and collaboration in medical AI.

"The greatest discovery is to find the profound in the simple."
- Alzheimer's Research Institute

Meet Our Dedicated Team

Behind the Alzheimer Disease Prediction system is a multidisciplinary team of experts committed to making a difference in medical diagnostics. Our diverse backgrounds in AI, neurology, data science, and software engineering converge to create a robust and reliable solution.

Dr. Anya Sharma
Lead AI Scientist
Specializes in deep learning models for medical image analysis and diagnostic AI.

Dr. Ben Carter
Clinical Lead
A seasoned neurologist with extensive experience in neurodegenerative diseases.

Sarah Lee, M.Sc.
Data Engineer
Designs and maintains secure and efficient data pipelines for MRI and clinical data.

Michael Wong, Ph.D.
Research Associate
Focuses on advanced image processing techniques and feature extraction for AI models.

Technology & Methodology

Our system leverages state-of-the-art deep learning architectures, including Convolutional Neural Networks (CNNs), optimized for analyzing complex patterns in MRI brain scans. We adhere to rigorous data privacy protocols, ensuring all patient data is handled with the highest level of security and anonymity.

How does the AI model work? ~

What measures are taken for data privacy? ~

What is the scientific basis of your predictions? ~

Frequently Asked Questions

How accurate is the Alzheimer Disease Prediction system? ~

What types of MRI data are supported? ~

Is patient data handled securely? ~

Can the system be integrated with existing EMR systems? ~

How frequently is the prediction model updated? ~

Get in Touch with Us

Have questions about the project, interested in collaboration, or need support? Reach out to our team.

Your Name:

Your Email:

Your Phone Number:

Your Message:

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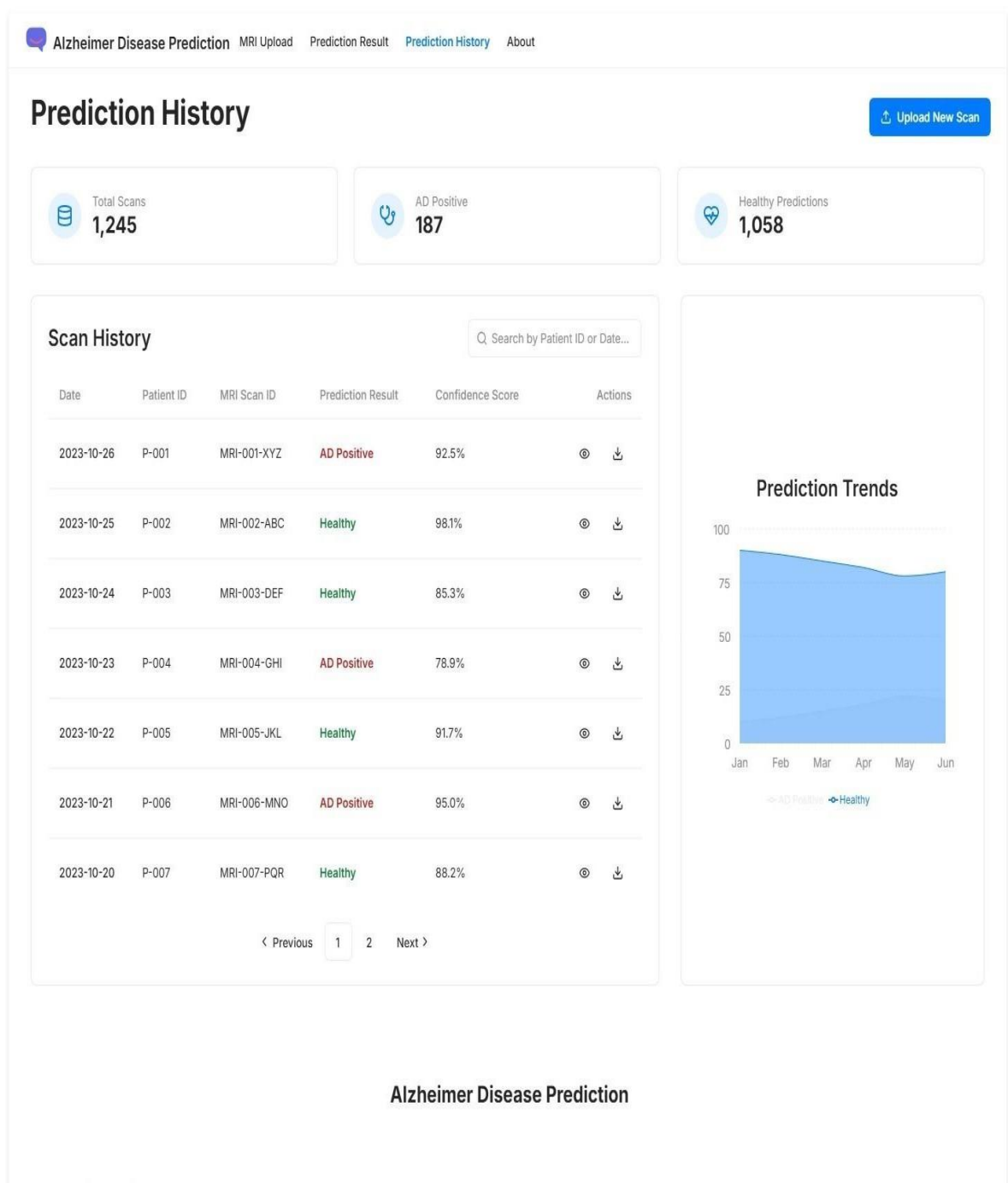
Our Location & Contact

123 AI Boulevard, Neuro City, NW 78901, USA

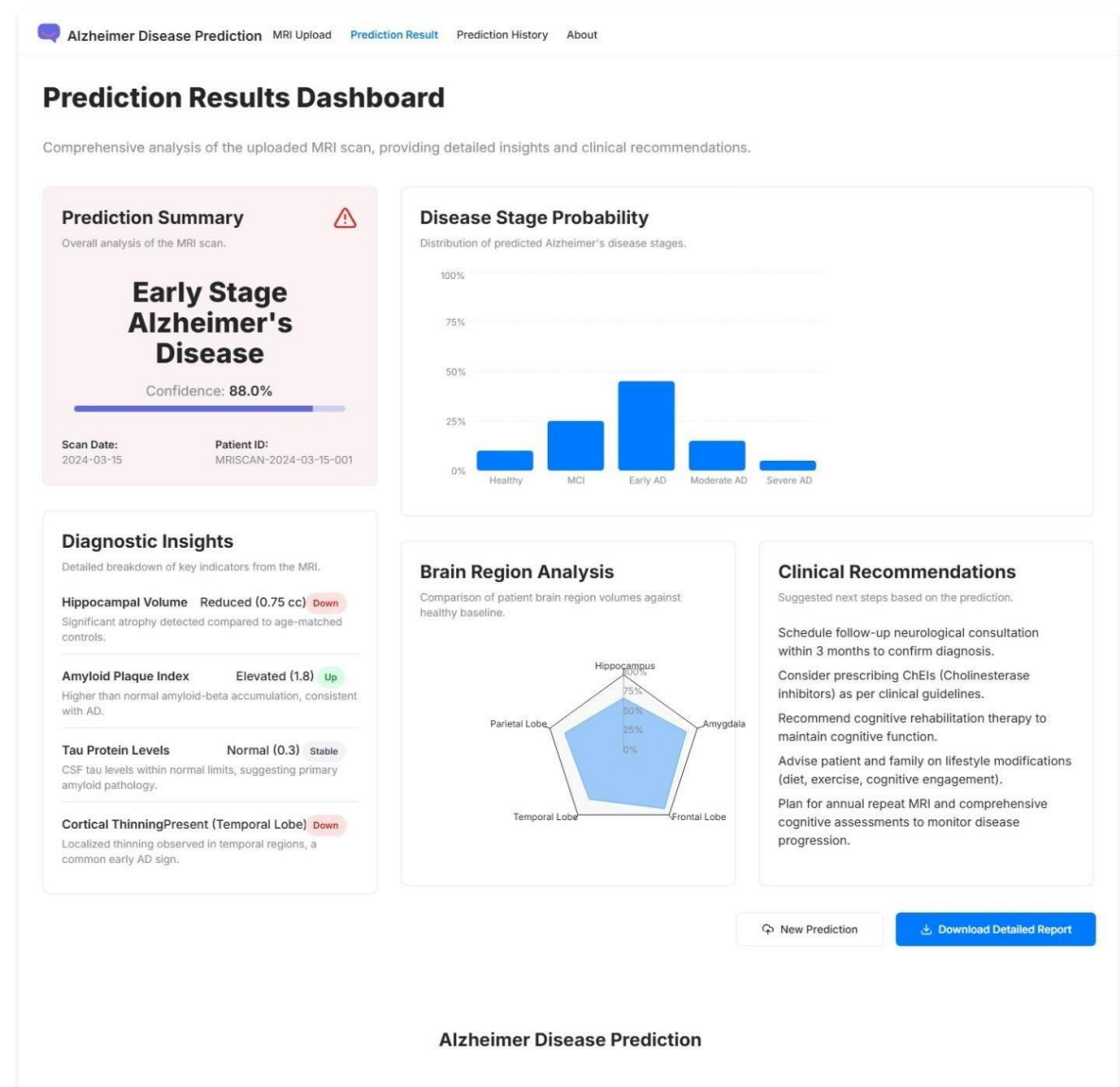
info@alzheimerprediction.com

+1 (555) 123-4567

History:



Results:



17. Conclusion:

This project presents a CNN-based system for the early prediction of Alzheimer’s Disease using MRI scans, aiming to improve diagnostic accuracy and reduce subjectivity. Automating image analysis minimises human error and speeds up the diagnosis process. The integration of a user-friendly interface ensures accessibility for healthcare professionals and supports practical clinical use. Visual explanations of predictions enhance trust and understanding. Overall, the system bridges the gap between AI technology and healthcare, contributing to timely and reliable AD detection.

18. References:

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19. Plagiarism Report

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